

Punnam Yuva Keerthana Venkata Gowrish

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EDUCATION

Indian Institute of Technology Bhubaneswar	8.98/10
<i>B.Tech in Electronics and Communication Engineering</i>	2024 – 2028
Sri Chaitanya Junior College – Intermediate (12th)	98.1%
<i>Board of Intermediate Education</i>	2024
Sri Chaitanya Techno School – Secondary School (10th)	572/600
<i>Board of Secondary Education, Telangana</i>	2022

TECHNICAL SKILLS

HDL/RTL: Verilog HDL, SystemVerilog (Basics), RTL Design, FSM Design, Digital Logic Design, RTL Verification
VLSI Concepts: CMOS Fundamentals, Computer Architecture, Logic Synthesis, Static Timing Analysis (STA), Physical Design Flow (RTL-to-GDSII)
Digital Digital: RISC-V (RV32I), Pipelined Processor Design, UART, Floating-Point Arithmetic, Communication Protocols, FPGA Design
EDA Tools: Vivado, Yosys, OpenSTA, GTKWave, ModelSim, VS Code
Programming: C, C++, Python, RISC-V Assembly, TCL (Basics)

PROJECTS

- RV32I RISC-V Processor Family** | *RTL Design Project* Jan 2026 – Mar 2026
- Designed two 32-bit RISC-V (RV32I) processors in Verilog HDL — a single-cycle architecture and a 5-stage pipelined architecture (IF, ID, EX, MEM, WB) — supporting 35+ RV32I instructions.
 - Enforced pipeline registers, data forwarding, and a hazard detection unit in the pipelined design to resolve data and control hazards and improve instruction-level parallelism.
 - Executed a comprehensive testbench suite covering 40+ instruction sequences; validated functional correctness against the official RISC-V compliance test suite with zero mismatches. Synthesized on Xilinx Basys3 FPGA via Vivado.
- BFloat16 Floating-Point Arithmetic Unit** | *Digital Design Project* May 2026 – Jun 2026
- Developed a 16-bit BFloat16 floating-point arithmetic unit in Verilog with a 2-bit operation-select input with correct rounding and special case handling (NaN, Inf, zero).
 - Structured the design into seven RTL stages to handle mantissa alignment, computation, normalization, and result repacking. Covered edge cases, validating output against expected results across multiple test vectors in simulation.
 - Synthesized the processor onto FPGA hardware with LED-based output monitoring for real-time result verification.
- UART Communication System** | *Communication Protocol Project* Oct 2025 – Dec 2025
- Designed a complete UART communication system modular RTL architecture, implementing the standard 8N1 frame format (8 data bits, no parity, 1 stop bit) at 9600 baud, with FSM-based transmitter and receiver modules.
 - Implemented two data-flow configurations: a normal mode following TX → RX → process → dataout, and a bidirectional mode following TX → RX → process → TX again → dataout, where received data is processed and re-transmitted to confirm round-trip communication integrity.
 - Built a dedicated baud rate generator module to derive accurate 9600 baud bit timing from the Basys3's 100 MHz onboard clock, using separate TX/RX baud tick generation.
- FPGA Digital Clock** | *FPGA Implementation* Aug 2025
- Operated a real-time 24-hour digital clock in Verilog with hours, minutes, and seconds counters driven by a 100 MHz on-board clock.
 - Displayed output on 7-segment display of Basys3 board using multiplexed display logic.

RELEVANT COURSEWORK

Digital Electronics, VLSI Design, Computer Architecture, Digital Signal Processing, Microprocessors & Microcontrollers, Signals & Systems, Electronic Devices & Circuits

ACHIEVEMENTS & ACTIVITIES

- Successfully solved all HDLBits exercises, demonstrating proficiency in Verilog HDL, RTL Design, Digital Logic Design, FSM Design, and Hardware Modeling.